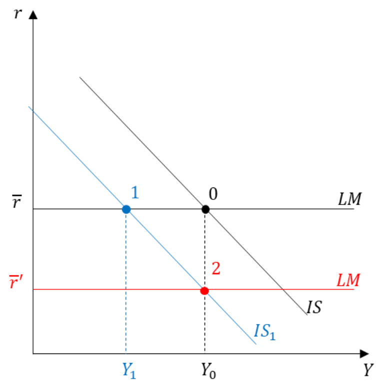
Question 1 (7.5 points)

The economy of country MACRO is described by an extend IS-LM model and it is in a short run equilibrium. Suppose that uncertainty due to the Middle East war increases the risk aversion among investors.

a) Describe the effects of this change on the equilibrium level of output and interest rates. How does the composition of the demand change in the new equilibrium? What about national saving and the real money demand? Explain economically and represent graphically.

b) Suppose that the central bank wants to return income to the initial level. Which policy intervention should the central bank implement? Compare the composition of the aggregate demand after the central bank intervention with the one prevailing before the change in risk aversion.

Possible answer



a) With an increase in the risk aversion of investors, the borrowing rate increases, investment declines and the demand is lower: the IS curve shifts to the left. As the Central Bank operates to keep constant the interest rate (flat LM), the economy is now at point 1, with lower GDP and the same interest rate. In equilibrium, consumption is reduced because of lower income, investment is lower too for both the decrease in income and the increase in the risk premium, and G is constant by assumption. Moreover, private saving decreases because of the reduction in income; public saving is constant and national saving thus declines.

Finally, as income decreases, the demand for money decreases too: to keep the interest rate constant the Central Bank must reduce the supply of money accordingly.

b) In order to increase income and restore the initial output level, Y, the Central Bank has to reduce the nominal interest rate by implementing an expansionary monetary policy. By buying bonds, on the bond market there will be a higher demand for bonds, their price will increase and therefore, given the face value, the nominal interest rate will decrease. The lower interest rate will make investment less expensive, will increase the aggregate demand and through the multiplier the income level. The LM curve moves downward up to crossing the IS' curve in point 2. In this new equilibrium, Y is unchanged while r is lower. As a result, by comparing the final equilibrium (2) with the initial one (0) we can conclude that since consumption has not changed (because Y is unchanged) and G is exogenous, the decrease in the interest rate must stimulate investment exactly enough to offset completely the downward effect of higher risk.

Question 2 (7.5 points)

Two BIEM students are discussing about the effects of a tax on bank deposits. Student 1 says *"A tax on bank deposits certainly reduces the money supply but it does not necessarily influence the interest rate: it depends on the behavior of the central bank"*. Student 2 replies: "*No, I don't agree. A tax on deposits reduces the demand for money and, consequently, the interest rate"*. Which of the two students is right? Justify your answer based on the relevant economic theory.

Possible answer

A tax on bank deposits "eats out" a chunk of individuals' deposits, reducing the appeal of deposits compared to currency. Agents will keep a higher fraction of their money demand (that does not change) as currency and they will deposit less on their bank accounts. In other words, the tax reduces the demand for deposits and increases the demand for currency (c increases). As a result, the money multiplier effect that takes place through deposits is decreased in magnitude and the aggregate money supply shrinks. Banks will have few resources to buy bonds or to make loans and they inject less liquidity in the economy. This puts upward pressure on the interest rate, but the ultimate effect on the interest rate depends on the behavior of the central bank. If the CB wishes to keep the supply of central bank money H fixed, then the nominal interest rate will increase. Instead, if it wishes to keep the interest rate fixed to its desired level or to keep constant the amount of aggregate money M supplied, it will have to conduct expansionary OMO, to increase the monetary base (H) and the money supply to shift it back to its initial level. Thus, student 1 is right. Student 2 is wrong since the money demand does not change: the demand for money depends only on the level of transactions in the economy and on the opportunity cost of money with respect to bonds.

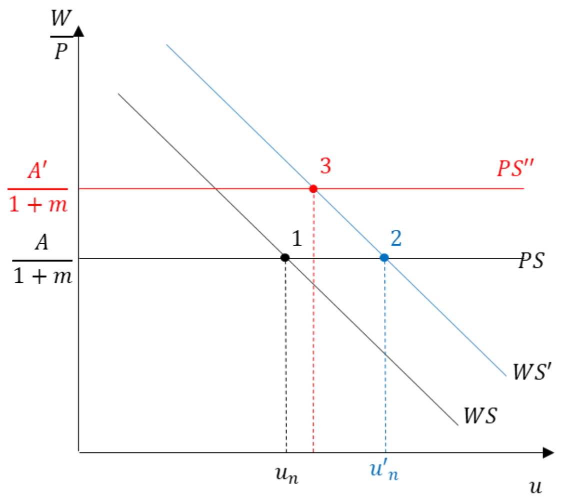
Question 3 (7.5 points)

In the labor market of country Gamma the unemployment rate is at its natural level. Suppose the government implements a reform that increases workers protection by increasing firms firing costs and by providing greater income support for workers who lose their jobs.

a) After having defined the natural unemployment rate, using the WS-PS model, explain how the natural unemployment rate and the real wage vary in the medium term. Provide a graphical representation and the economic intuition.

b) Suppose the government decides to combine the previous reform with a strong investment in education and training for workers, which increases their productivity by 25%. How does the labor market equilibrium change in the medium run? What about the potential output? Explain economically.

Solution:



a) The natural unemployment rate is the unemployment rate that the economy reaches when expectations are correct, in other words when $\pi_t = \pi_t^e$. It is the unemployment rate prevailing in the medium run and it depends only on structural factors.
If the government increases worker protection, z increases, leading to a shift upward of the WS curve to WS' in the graph. The PS curve does not change. This leads to the equilibrium point 2 in the graph, in which the real wage is the same but the natural unemployment rate is higher.
In terms of economic intuition, this means that workers now have more bargaining power and will ask for higher nominal wages, but since firms face the same markup and the same level of worker productivity, the real wage will not change (thus producers will charge higher prices) and this will result in fewer workers being hired and the unemployment rate rising. Labor market conditions should worsen to induce workers to accept the same real wage paid by firms despite higher workers' protection: the natural rate of unemployment increases (point 2).

b) Investment in education raises worker productivity, so A will increase. This shifts the PS curve upward, crossing the WS curve at a higher level of real wage and a lower level of natural unemployment. Thus, the potential output will be higher. However, without further information about the size of the two shocks (the increase in z and in A) we cannot say if the natural unemployment rate will be the same, will be higher or will be lower than the initial one. Economically, since workers are now more productive, and firms produce more output using the same amount of labor, firms set lower prices for given nominal wages, and the real wage paid to workers increase.

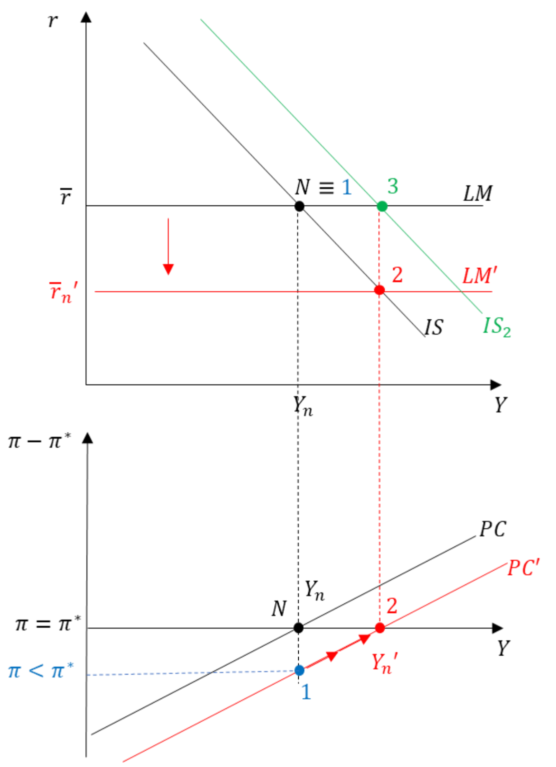
Question 4 (7.5 points)

The macroeconomic system of country ABC is described by a standard IS-LM-PC model. Assume that expectations are well anchored to the central bank target and that the country is in its medium run equilibrium, denoted by N.

a) Suppose that the government decides to pass a labor-market reform, which reduces labor protection. What happens in the short and in the medium run if the Central Bank wants to keep inflation at the announced target?

b) Suppose now that, after the labor reform described in part (a), the government decides to increase government spending G. Can a higher G help the economy reach its new medium-run equilibrium?

Solution:



a) If the reform of the labor market is passed, workers have less bargaining power and they are willing to accept lower wages. The WS curve in the labor market shifts down. As the wage paid by firms is lower, their marginal cost is lower and they set lower prices. The real wage remains the same, but the natural rate of unemployment declines; this corresponds to an increase in potential output, Y_n

Graphically, this policy shifts the PC curve to the right, while the IS and the LM aren't affected. Point N, then, is no longer a medium run equilibrium and GDP will be now lower than the new natural level Y_n'. In the graph, to remark that this intersection between the IS and the LM is a short run equilibrium, the point is relabeled "1".

As $Y_1 < Y_n'$ and $u_1 > u_n'$, workers are willing to accept lower wage growth and the growth in prices will be lower: the inflation rate π decreases below the Central Bank target.

In order to restore the target level of the inflation rate, the Central Bank must intervene with an expansionary monetary policy, buying bonds on the secondary market, to reduce the interest rate, increasing the level of GDP at the new medium run level.

b) Yes! In fact, an increase in public expenditure G increases demand and then the equilibrium level of GDP in the goods market. Graphically, the IS shifts right. If this shift is big enough, the economy can move to the new medium run equilibrium, or, alternatively, the magnitude of the reduction in the policy real rate by the Central Bank can be smaller than in case a). [point 3 in the graph]